

Validation and Testing

To ensure the reliability of the final Power BI solution, I performed a series of validation checks across the Power Query transformations, data model, DAX measures, and dashboard interactions.

Power Query Validation

I verified that the transformation steps produced the expected outputs. After removing duplicates, the Forecast dataset reduced from 246,477 rows to 240,348 rows, while the Actual dataset reduced to 53,276 unique Item-Country-Period combinations following deduplication and aggregation.

The Item field was validated to ensure that all values had been successfully converted to text format, eliminating potential merge failures caused by mixed data types. I also confirmed that no numeric Item values retained the ".0" suffix after standardisation.

The final Fact_SIOP table was checked to ensure that forecast columns contained only numeric values with no transformation errors. Forecast null values were replaced appropriately, and Actual CAPEX quantities remained zero throughout the dataset, consistent with the source data. I also confirmed that the AsOfPeriod field was no longer present in the final fact table, indicating that the snapshot transformation had been completed successfully.

Data Model Validation

The data model was validated to confirm that relationships behaved as intended.

The relationships between Fact_SIOP and both Dim_Period and Dim_Country were active, many-to-one, and configured with single-direction filtering. The Dim_LagSelector table remained intentionally disconnected to support slicer-driven DAX logic without directly filtering the fact table.

Dimension tables were also verified to contain the expected values, including 15 period records, 4 country records, and 3 lag labels.

Lag Slicer Validation

I tested the Lag Selector extensively to confirm that DAX measures returned the correct forecast values.

Selecting Resultant, Lag 0, and Lag 1 correctly switched the measures to their corresponding forecast columns. Clearing the slicer defaulted the report back to Resultant through the SELECTEDVALUE logic.

I also verified that Total Forecast Quantity always matched the sum of Revenue, Non-Revenue, and CAPEX forecast measures for the selected lag.

Forecast Accuracy and Bias Validation

Forecast Accuracy and Bias measures were tested under multiple scenarios.

For periods where actual data existed, Forecast Accuracy returned meaningful values. For future periods with no actual quantities, the measures correctly returned BLANK instead of displaying misleading zero values.

Bias measures behaved as expected, with positive values indicating over-forecasting and negative values indicating under-forecasting.

Since CAPEX actual quantities were zero and Non-Revenue actuals were minimal, combined accuracy measures closely aligned with Revenue-only accuracy calculations.

Dashboard Behaviour Validation

Interactive dashboard functionality was tested to ensure a consistent user experience.

Country filters updated all visuals appropriately, lag selections propagated across report pages, and period slicers correctly filtered trend visuals.

The Forecast Performance Summary responded dynamically to user selections, while future periods displayed blank accuracy values rather than artificial percentages.

Duplicate Removal Validation

Particular attention was given to validating duplicate handling because this represented the largest data quality issue in the project.

The Forecast dataset reduced from 246,477 rows to 240,348 rows after duplicate removal. The Actual dataset reduced from 500,000 rows to 57,946 rows after deduplication and finally to 53,276 rows following aggregation.

Finally, I verified that no row inflation occurred during joins by ensuring that the total Actual Revenue Quantity in Fact_SIOP matched exactly with the corresponding total in the cleaned Actual staging table. This validation was especially important because any fan-out during merging would have resulted in overstated actual quantities and misleading forecast performance metrics.